MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP BCA (HONOURS)

FIRST SEMESTER EXAMINATION

(2024 ADMISSION ONWARDS)

MG1SECBCA100: FUNDAMENTALS OF PROGRAMMING USING C

MODEL QUESTION PAPER

Duration: 2hrs Maximum Marks: 70

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A:

Very Short Answer Questions.

Answer All Questions. Each question carries 2 marks.

1. What are the steps involved in the program development life cycle? [K][1]
2. Identify the types of language translators used in programming and briefly explain each. [K][1]
3. How does an expression differ from a statement in C? Provide examples. [U][2]
4. Write a do-while loop in C to repeatedly prompt the user until they enter a positive number. [An][3]
5. What are the ASCII value ranges for uppercase and lowercase characters in C? [K][2]
6. What is the purpose of the continue statement in C? [K][3]
7. Describe how you would compute the total cost of items stored in an array of prices. [A][4]
8. Analyze the output of the following code snippet and explain how printf() formats the output. float num = 45.6789; printf("%.2f", num); [An][3]
9. Demonstrate how to declare and call a function in C that returns an integer. [A][4]
10. What is the primary difference between a struct and a union in C? [U][4]

(10 X 2 = 20 Marks)

Part B:

Short Answer Questions.

Answer any 5 out of 7 Questions. Each question carries 6 marks

11. Outline a systematic approach to debugging a C program, and explain how debugging tools help, identify syntax, logical, and runtime errors. [U][1]
12. Explain operator precedence in C with an example. How does it affect expression evaluation? [U][2]
13. Why are delimiters important in C programming? Provide examples to illustrate their role. [K][2]
14. Write a C program to read a user's name and age, and display them using both printf and puts. Discuss the advantages of each function in this context. [A][3]
15. Write a recursive function in C to compute the nth Fibonacci number. [A][4]
16. Analyse the statement: "The switch statement is less flexible but more efficient than if-else." Provide examples and reasoning to support your analysis. [An][3]
17. Implement a recursive function to calculate the nth Fibonacci number. [A][4]

(5 X 6 = 30 Marks)

Part C:

Essay Questions.

Answer any 2 out of 3 Questions. Each question carries 10 marks

18. Compare the use of if-else statements and the if-else-if ladder in C. Provide examples to show when one approach may be preferred over the other. [An][3]
19. Describe the different storage classes in C (e.g., auto, static, extern, register) with examples. Explain how they influence variable behaviour. [U][4]
20. Define the following terms in the context of functions in C:

    a) Function Declaration b) Function Definition c) Function Call

    d) Return Type.

    After defining each, explain the differences between a function declaration and a function definition with examples. [U][4]

(2 X 10 = 20 Marks)